Universidad de los Andes

AUSELIA

Acoustic Edge-AI Detection of Crop Water Stress with a Custom RISC-V Processor

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A drying plant makes ultrasonic clicks when the water column in its xylem breaks. The Auselia node hears these clicks and flags water stress on its own low-power RISC-V chip.

01 MOTIVATION

Hearing drought before it shows

Chile is a leader in premium fruit and wine. Climate shifts and water scarcity put this at risk. Growers need to find water stress before the plant shows damage.

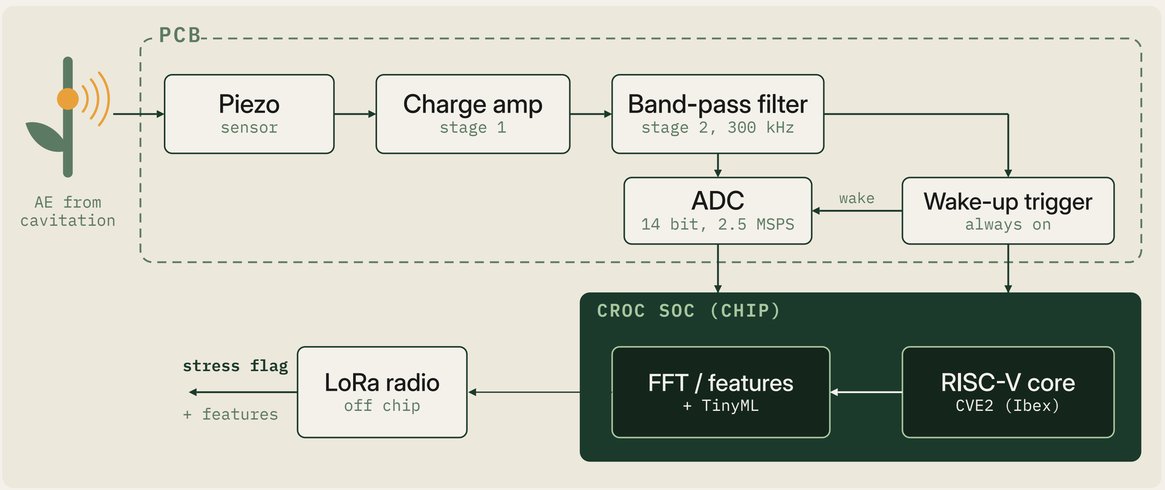
In a plant that loses water, the sap in the xylem is under high tension. The water column can break (cavitation). Each break releases a short ultrasonic pulse. These acoustic emissions (AE) are in the 100 kHz to 1 MHz band and come before visible symptoms [1, 2].



02 SYSTEM

Auselia node architecture

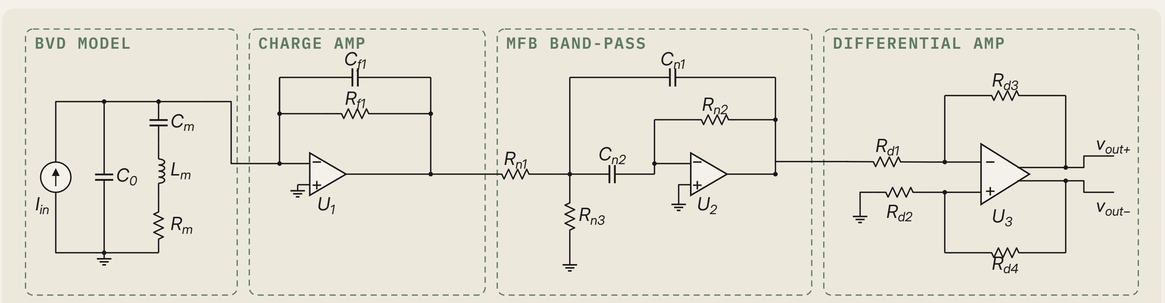
Three parts: a piezoelectric analog front end, a custom low-power RISC-V processor and on-device TinyML.



03 ANALOG FRONT END

Three stages from charge to volts

We model the transducer with a Butterworth-Van Dyke (BVD) network [3]. Three stages share the gain and the filtering [4, 5].



1. **Charge integrator (OPA1642).** Charge to voltage, high input impedance, first gain.
2. **MFB band-pass, 2nd order, 300 kHz.** Removes DC drift and out-of-band noise.
3. **Differential output (THS4531).** Balanced gain, common-mode rejection, ADC range.

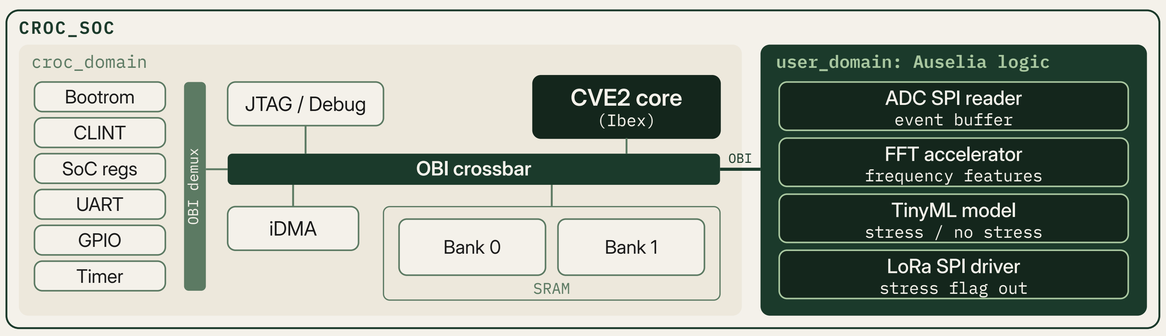
LTspice results. Cavitation pulse plus thermal noise, tractor vibration [6], bird-scaring cannons [7], wind in foliage [8] and drip irrigation [9].

300 kHz	≈54 dB	>40 dB	12 mVpp
band-pass centre	combined gain	attenuation below 10 kHz	clean output under full noise

04 DIGITAL

Open RISC-V SoC on Croc

Croc is an open-source RISC-V microcontroller with an open RTL-to-layout flow [10]. We add the Auselia logic in its user domain, so the open flow and the low pin count stay the same. The core sleeps until the wake-up trigger fires. This keeps the average power low [11, 12].



05 EXPERIMENT

Bench dehydration of a blueberry

One potted Emerald blueberry dried on the lab bench. A piezoelectric sensor on the stem fed the charge amplifier. An oscilloscope recorded each triggered event. We first characterized the sensor with Hsu-Nielsen pencil-lead breaks.

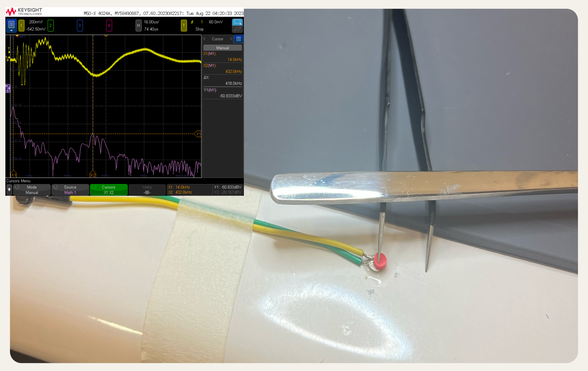


Figure 1: Hsu-Nielsen characterization of the piezoelectric sensor

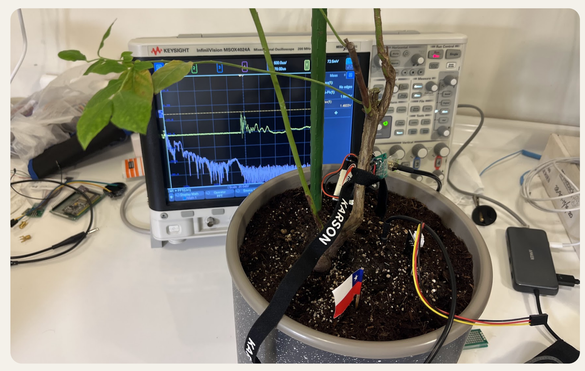
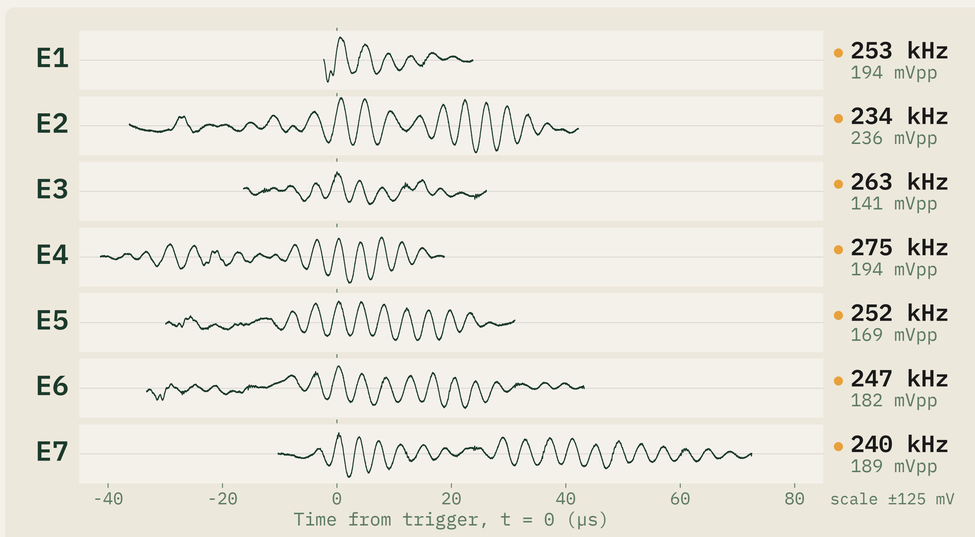


Figure 2: Emerald blueberry with the sensor on the stem and oscilloscope capturing sensor output.

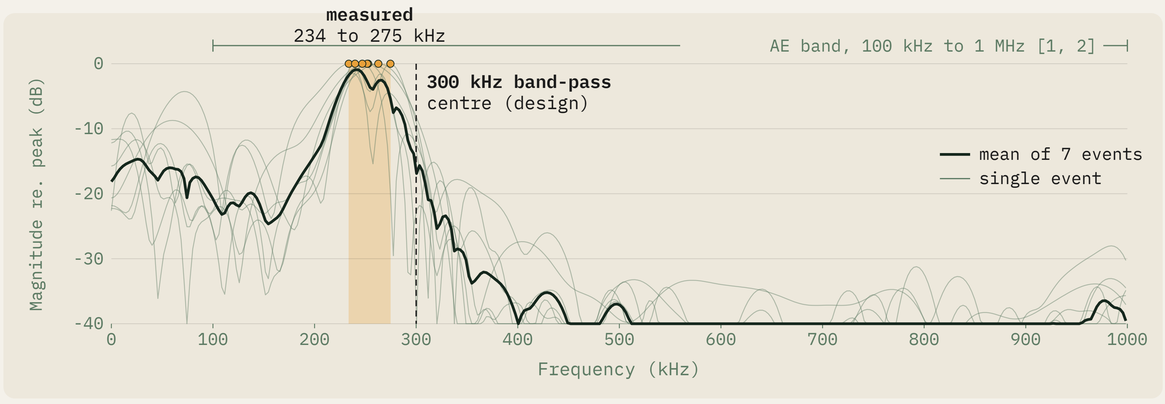
06 RESULTS

First cavitation events not verified

Seven events recorded. All seven show a short burst of ringing with the same dominant frequency.



Waveforms. Charge-amplifier output, aligned at the scope trigger. The right column gives the peak frequency and the peak-to-peak amplitude of each event.



Spectra. Magnitude of each event relative to its own peak. Amber marks the measured peaks: 234 to 275 kHz, mean 252 ± 14 kHz. Amplitude 141 to 236 mVpp.

07 DISCUSSION

What the first events tell us

The band is right, the centre is low. The energy sits at 234 to 275 kHz. This is inside the expected AE band but below the 300 kHz filter centre. The next front-end revision will move the band-pass centre toward 250 kHz or widen the pass band.

The ADC is fast enough. At 2.5 MSPS the planned ADC covers the signal up to 1.25 MHz.

Limits. One plant on the bench. Short records (12 to 38 kHz spectral resolution). No independent check yet that each event is cavitation.

NEXT 1
Build a labeled dataset. Pair each recording with plant water status and climate data.

NEXT 2
Put the front end, the Croc core and the TinyML model on one chip.

NEXT 3
Build and validate an end-to-end field prototype.

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